The CAB inputs include the following:

- four wheel speed sensors
- brake lamp switch
- ignition switch
- battery voltage
- diagnostic communication
- G switch (acceleration switch)

The CAB outputs include the following:

- six valve/solenoid drivers
- pump/motor actuation
- ABS warning indicator control
- diagnostic communication

3.1.5 HYDRAULIC CONTROL UNIT

The hydraulic control unit (HCU) contains the valve block assembly, three accumulators, and the pump/motor assembly.

Valve Block Assembly: The valve block assembly contains 6 valves with three inlet valves and three outlet valves. The inlet valves are spring-loaded in the open position and the outlet valves are spring loaded in the closed position. During an antilock stop, these valves are cycled to maintain the proper slip ratio for each channel. If a wheel locks, the inlet valve is closed to prevent any further pressure increase. Then the outlet valve is opened to release the pressure to the accumulators until the wheel is no longer slipping. Once the wheel is no longer slipping, the outlet valve is closed and the inlet valve is opened to reapply pressure. If the wheel is decelerating within its predetermined limits (proper slip ratio), both valves will close to hold the pressure constant.

Pump/Motor Assembly: The pump/motor assembly provides the extra amount of fluid needed during antilock braking. The pump is supplied fluid that is released to the accumulators when the outlet valve is opened during an antilock stop. The pump is also used to drain the accumulator circuits after the antilock stop is complete. The pump is operated by an integral electric motor. This motor is controlled by the CAB. The CAB turns on the motor when an antilock stop is detected. The pump continues to run during the antilock stop and is turned off approximately 3-5 seconds after the stop is complete. The pump mechanism consists of two opposing pistons operated by an eccentric cam. One piston supplies the primary hydraulic circuit. The opposing piston supplies the secondary hydraulic circuit. In operation, one piston draws fluid from the accumulators. The opposing piston pumps fluid to the valve body solenoids. The CAB monitors the pump/motor operation.

Accumulators: The accumulators provide temporary fluid storage during an antilock stop and are drained by the pump/motor.

3.1.6 RELAYS/SWITCHES

Relays: The ABS Relay is located in the power distribution center (PDC). When the relay is de-energized, it supplies a ground to turn on the ABS warning indicator. The ABS relay is controlled by the CAB. The CAB energizes the relay when it receives an ignition input signal.

3.1.7 SENSORS

Wheel Speed Sensors and Tone Wheels: One wheel speed sensor (WSS is located at each wheel and sends a small AC signal to the Controller Antilock Brake (CAB). This voltage is generated by magnetic induction when a toothed sensor ring (tone wheel) passes by a stationary magnetic sensor (wheel speed sensor). The CAB converts the AC signals into digital signals for each wheel.

The front wheel sensor is attached to a boss in the steering knuckle. The tone wheel is an integral part of the front axle shaft. The rear speed sensor is mounted in the caliper adapter plate (rear disc only) and the rear tone wheel is an integral part of the rear rotor hub. The front wheel speed sensor air gap is NOT adjustable. The correct front wheel speed sensor air gap is from 0.40mm to 1.3mm (0.016″ to 0.051″). Preferred rear sensor air gap for drum brake systems is 1.1mm (0.043″). Acceptable air gap range is 0.92mm to 1.275mm (0.036″ to 0.050″). All wheel speed sensors have a resistance between 900 and 1300 ohms.

Correct antilock system operation is dependent on correct wheel speed signals from the wheel speed sensors. The vehicle's wheels and tires should all be the same size and type. In addition, the tires should be inflated to the recommended pressures for optimum system operation. Variations in wheel and tire size or significant variations in inflation pressure can produce inaccurate wheel speed signals; however, the system will continue to function when using the mini-spare. When driven over rough road surfaces, the rear wheel speed sensor signals may be erratic and cause a false trouble code.

G (Acceleration) Switch: The CAB monitors the acceleration switch at all times. The switch assembly contains three mercury switches that monitor vehicle deceleration rates (G-force). Sudden, rapid changes in vehicle and wheel deceleration rate trigger the switch, sending a signal to the CAB. The switch assembly responds to three deceleration rates; two for forward braking and one for rearward braking.